R6.2: Native ALT Evidence and Hardware Service Boundary

Separate semantic track. The strict CPU manuscript retains native tie semantics and map-equivalent aggregates. This appendix evaluates original-format ALT_BUCKET_FP32_V1 at fixed shift20. Whole-pair FP32 wave updates can select different congestion values from STRICT. Compact/direct implementations remain offline-only; native qualification is not inferred.

Frozen holdout and scope. LSTM seed2 was selected before timing independently of shift/cache tuning: two layers, width120, 4,166,724 final RR nodes, 40,142,314 edges and 40 samples, within unchanged $N < 2^{23}$ admission. Stock, Flat, Pipeline, original ALT CPU and live ALT CUDA use fixed netlist/placement/RRG inputs on one H100 worker and eight physical cores. Three independent processes rotate arm order. E2 is the whole native lookahead constructor; Process includes RR XML loading, initialization, routing, checks and STA. Prior input generation is excluded; filesystem cache is not flushed.

Qualification. Live CUDA compares every sample pair with a same-input CPU reference in a diagnostic run. Formal runs exclude capture and that verification. Every route retains its identity through independent FULL/STA, zero overuse and a 5% CPD/wirelength gate against same-cohort Flat and stock. The no-fanin audit explains isolated outer-boundary CHANZ using exact channel-coordinate rules and final degrees; it does not certify all RRG invariants. No graph or checker was changed. Stock E2 has 0.01-second log precision; other E2 values use the high-resolution constructor timer. Timer formatting is not measurement accuracy.

Table 1: LSTM native seconds, mean ± sample SD ($n = 3$), one worker.

Method	E2	Process
Stock	65.413 ± 4.066	137.395 ± 4.839
Flat	9.909 ± 0.578	81.545 ± 1.742
Pipeline	9.907 ± 0.563	81.727 ± 1.986
ALT CPU	9.781 ± 0.179	81.246 ± 1.204
ALT CUDA	3.089 ± 0.071	74.703 ± 1.100

Table 2: LSTM control/CUDA paired speedup, mean ± sample SD ($n = 3$); no ratios of medians here.

Control	E2	Process
Stock	21.182 ± 1.342	1.839 ± 0.049
Flat	3.208 ± 0.165	1.092 ± 0.007
Pipeline	3.207 ± 0.147	1.094 ± 0.010
ALT CPU	3.167 ± 0.071	1.088 ± 0.000122

The held-out ALT/Flat maps differ in 0 delay words, 23111 congestion words and 0 missing states across 459684 entries. Original ALT CPU/CUDA input-prefix and full-output hashes agree. These map counts are not per-sample difference counts.

One conversion optimization. A single non-formal BNN profile measures 0.812 s in the dense mapping/job/cost construction interval, versus 0.619 s in CSR construction/allocation and 0.119 s in H2D. The sole candidate pre-indexes IPIN IDs and fills the unchanged original dense buffer. Both old cases pass affected CPU/CUDA qualifications; complete input-prefix and output SHA256 checks retain original bytes. The following measurements compare only live CUDA before/after this one bridge change on the same worker. They do not transfer optimized native qualification or timings to LSTM.

Table 3: Original/candidate paired speedup, mean ± sample SD ($n = 3$).

Case	Scope	Speedup
s38417	E2	0.983 ± 0.085
s38417	Process	1.014 ± 0.032
bnn	E2	1.082 ± 0.037
bnn	Process	1.009 ± 0.019

BNN E2 improves in all three pairs, but neither case has a consistently positive Process effect. The small-case E2 result also has mixed direction. The candidate remains an isolated experiment; defaults are unchanged and no second optimization is attempted.

Earlier cohort and Pipeline boundary. Recomputed from R6.1 raw repeats, BNN Flat/CUDA paired E2 speedup is $3.274 \pm 0.222\times$, and Process speedup is $1.091 \pm 0.011\times$. The earlier report used ratios of medians, $3.348\times$ and $1.095\times$. These statistics are not interchangeable. CUDA is slower on s38417; qualified ALT CPU builds faster, while stock has the fastest Process. R6.2 Pipeline supplements for s38417 and BNN pass map/QoR checks on a new worker; they do not support ratios to old-worker CUDA times. LSTM supplies the new same-worker Pipeline comparison.

System limit. In BNN R6.1 repeat1, E1 is 0.363794840 s of E2=3.104359912 s (11.72%). Keeping every other interval fixed, eliminating E1 gives at most $1.132744\times$ E2 or $1.005434\times$ Process speedup. This does not constrain conversion/preparation optimization. Stage partitions use one process; child medians are never summed. Historical E1/service ends before final ABI output handoff; E2 includes it.

Hardware stop. Read-only Vivado 2022.2 identifies 960 URAM, 2016 BRAM36 and 3 SLRs in U280. XRT finds no FPGA and platforminfo no installed platform. One 32-bit member per 4096×72 URAM word needs 1024 URAM for 16 MiB, exceeding the raw device; two members per word need 512 but do not establish ports or placement. Cache lookup/LRU mapping and random dependent RMW service remain unproved. Software hit counts are not hardware throughput. Proposal traffic alone implies 118.00 GB/s break-even and 147.50 GB/s with 20% time margin against the frozen warm compact-CUDA E1 target; other traffic is omitted. No achieved frequency or calibrated whole-pair HBM service is available. Status is UNRESOLVED_NO_SUPPORTED_MEMORY_EVIDENCE; FPGA superiority is false. No HLS/RTL/OOC or new replay was launched.

Evidence scope. Three runtime repeats of one holdout placement do not establish all-architecture or all-size generality. The strict CPU paper and this ALT appendix remain separate. Per-case/backend/shift/representation records, negative outcomes, source/object bindings and raw logs are archived in the R6.2 evidence snapshot. No automatic next hardware round follows.